

(TECH MANAGEMENT)

Organizing for Innovation Technology and Pattern Innovation

Invention :- Coming up with new idea or tech

Innovation :- Taking that idea and bringing it to the market

Types of Innovation (Schilling's Taxonomy)

①

Product Innovation vs Process Innovation

↳ changing what you sell

↳ changing how you make or deliver it

They go together :- Better process (Gigafactory) make better products (Cheaper EVs). You can't always separate them.

② Radical Innovation vs Incremental Innovation

↓

Fundamentally different: Breaks the mold. Often creates a whole new market.

↳ High risk, high reward

↳ Favours new entrants not incumbents

↓

Small improvements on what already exists

↳ Lower risk

↳ Helps incumbents maintain position

Example :- iPhone 15 camera upgrade.

Example :-

First iPhone (2007) -

nobody had a touch screen

smartphone. Tesla EVs -

entirely different from gas cars

Remember :- Radical is relative. What's radical for a small startup might be incremental for a tech giant. It depends on the firm's perspective.

Competence Enhancing vs Competence Destroying

↓
Builds on what you know:-

↓
Make old skills useless.

The new technology uses the skills and knowledge a firm already has. Existing players can adapt.

The new technology obsolesces existing skills. Even experienced firm starts from scratch.

Examples:-

Apple designs its own chips (M1, M2). This builds on their expertise in software + hardware integration. They're better at it than anyone else.

Examples:- EV drivetrains completely destroy the value of knowing how to build internal combustion engine parts. Toyota had decades of expertise in engines - now that expertise is becoming worthless.

Strategic Implication:-

If you're an incumbent (established company), competence-destroying innovations are terrifying. Your biggest strengths become irrelevant. New entrants (like Tesla) who never had those old competences can leapfrog you.

Architectural vs Component Innovation

↓
Redesign the whole system.

↓
Change one part.

You redesign how all the parts fit together. The system level logic changes entirely.

↓ You improve a single part of the system. The overall structure stays the same.

E.g.: Tesla entire EV platform software + hardware + even the

E.g.: Tesla improves battery cell chemistry (e.g.: 4680 cells). The car architecture is the same, but the battery is better - more range, cheaper cost.

air updates + dined sales model
all integrated together. No
one had connected all these
pieces together this way before.

Architectural Innovation is usually more disruptive than component innovation.

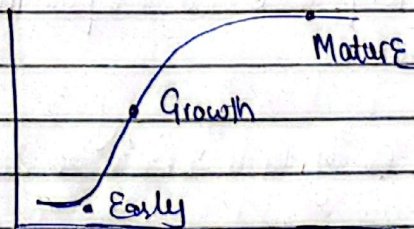
Patterns of technological change:-

follows recognizable, repeating patterns
that managers can use to predict and strategize.

- (i) Performance improve in an S-curve shape.
- (ii) Adoption by customers also follows an S-curve diffusion.
- (iii) Industries go through technology cycles (fledgling $\rightarrow$ dominant design $\rightarrow$ incremental).

The technology S-curve:-

Performance



① Early phase:- Slow phase

R&D investment/time.

Lots of R&D, small gains. like the first electric cars in 2005 - huge efforts, limited range. The tech is being figured out.

② Growth phase - Rapid Improvement:-

Scientists crack the key problems. Performance jumps fast. like EV battery range leaping from 150 km to 300 km in a few years.

③ Mature phase:- Diminishing Returns

You have hit the physical limits. Huge investment = tiny improvement. Going from 300km to 510km range takes as much R&D as going from 150 to 350km.

Strategic Implication:-

firms must know when to "jump curves" - abandoning a new tech & investing in the next S-curve.

Example:-

Nokia stayed on the feature phone S-curve too long. Apple jump to the smartphone S-curve. Nokia died. Apple thrived.

Limitation of S-curve:-

① descriptive, not predictive - you can explain the past - but hard to predict the future -

② Hard to know exactly where you are on the curve in real time -

③ External shocks (geopolitics, gov regulation) can disrupt the trajectory completely.

Technology Diffusion S-curve (Adoption):-

S-curve → who buys the technology and in what order - not about how good it is.

The 5-Adopter Groups:-

2.5%	13.5%	34%	34%	16.1%
Innovators	Early adopters	Early majority	late majority	laggards

Technology Cycles and Dominant design:-

Industry follows a predictable cycle:-

(i) Fluid phase:- Many different design compete. Nobody knows what will win. High variety, high experimentation.

(ii) Dominant design emerges:- One design wins. The market rebuilds around it. Everyone else either adopts it or exits.

(iii) Era of Incremental change:-

The dominant design is set. Innovation focuses on fine-tuning it. Competition shifts to cost, efficiency, brand.

Open vs closed Innovation

WHY Open can WIN:-

(1) Grows the whole ecosystem faster

(2) Accelerates adoption - more partners, more infrastructure.

(3) Builds goodwill and industry leadership

(4) Tesla: Opened patents → EV charging network expanded → Tesla Benefits

WHY closed can WIN:-

(1) Protects IP & competitive adv

(2) Controls the customer experience end-to-end.

(3) Competitors can't copy your core innovation

(4) Apple:- closed iOS = customers can't leave, pays premium forever.